From Fixed-Depth to Adaptive Prefetching: A Two-Phase Stream Buffer Study

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Abstract

This report describes a two-phase study of hardware prefetching via stream buffers. Phase 1 implements Jouppi's original fixed-depth FIFO stream buffer within a functional (timing-free) Pin-based cache simulator and establishes baseline sensitivity to buffer depth, stream count, and L1D associativity across three SPEC CPU2006 workloads. Phase 2 advances to an adaptive prefetch-depth policy modelled on Palacharla and Kessler's follow-up paper, adding a two-level cache hierarchy, a latency model, and a histogram-driven policy that learns stream-length distributions at runtime to choose prefetch depth dynamically. The adaptive design achieves 5.02× speedup on `libquantum` (vs. 5.83× for static next-line) while attaining 83% prefetch accuracy on `dealII` (vs. 78% for next-line) and reducing wasted bandwidth on irregular workloads by an order of magnitude relative to both the fixed-depth Phase 1 design and the next-line policy.

1 Introduction

The memory wall — the growing latency gap between the processor and DRAM — makes prefetching a critical hardware optimization. Stream buffers, introduced by Jouppi [1], are a lightweight mechanism that detects sequential access patterns and issues speculative fetches without requiring compiler or OS support. Their core weakness is that a fixed prefetch depth wastes bandwidth on non-sequential workloads while still failing to fully hide latency on very long sequential streams when latency is high.

Palacharla and Kessler [2] subsequently showed that the optimal depth is workload-specific and varies even within a single benchmark over time. Their key insight is that a histogram of observed stream lengths — gathered online during execution — can guide prefetch depth selection, making the prefetcher aggressive where it helps and conservative where it does not.

This project implements both designs as Intel Pin tools, evaluates them on `libquantum` (quantum circuit simulation, highly sequential), `hmmer` (hidden Markov model matching, highly irregular), and `dealII` (finite element analysis, mixed), and analyzes the resulting accuracy, coverage, and hardware cost trade-offs.

2 Phase 1: Fixed-Depth Stream Buffer

2.1 Design

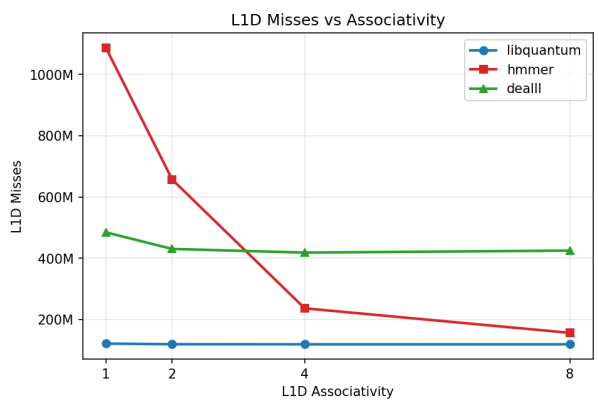
The Phase 1 simulator models a three-level hierarchy (L1I, L1D, L2, memory) where each level uses a common LRU set-associative cache base class. The stream buffer sits between L1D and L2 as an array of N stream descriptors, each holding a validity flag and a `head` — the block-aligned address of the oldest prefetched line in the FIFO. On every L1D miss the simulator first probes the stream buffer; only on a combined miss does it fall through to L2 and allocate a new stream.

Access. A hit occurs when the demand address matches any stream's `head`. The head advances by one block and a single new tail line is fetched from L2, maintaining exactly D lines in flight at all times.

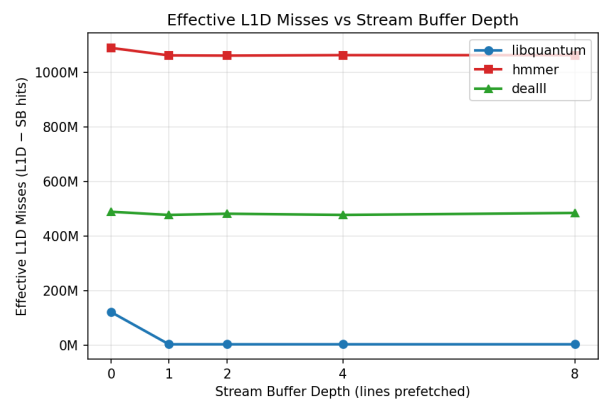
Allocation. On a full miss, the simulator selects a slot (preferring unused slots; falling back to round-robin eviction) and immediately issues D prefetch requests to L2 starting one block past the faulting address. The faulting line itself is fetched by the L1D miss handler.

Simulator limitation. The Phase 1 simulator is *functional*: it counts hits and misses but models no memory latency. As a consequence, depth has no effect on hit rate — the hit condition checks only whether the demand address matches `head`, with no notion of whether the prefetched line has had time to arrive. Depth only changes how many L2 prefetch requests are issued per allocation.

2.2 Phase 1 Results

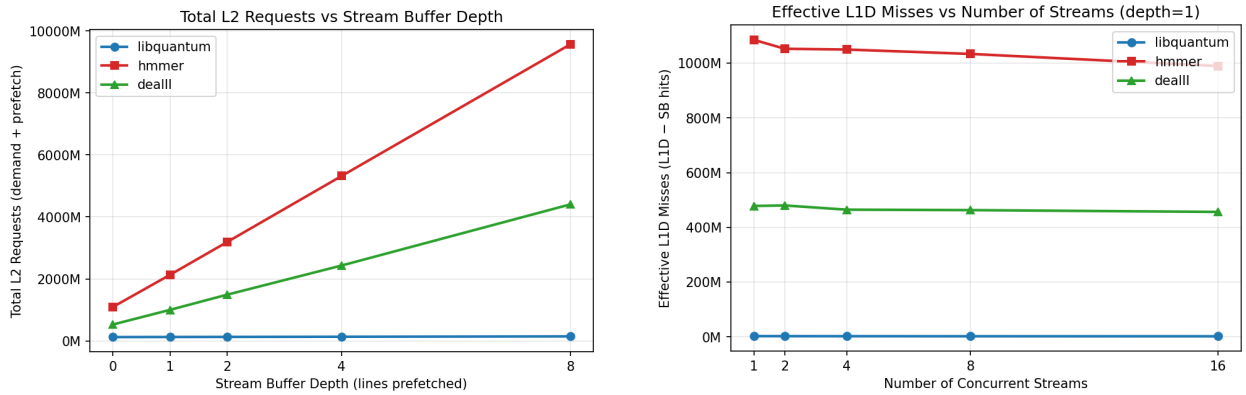


(a) L1D misses vs. associativity (no stream buffer).



(b) Effective L1D misses vs. stream buffer depth ($N = 1$).

Figure 1: Phase 1 baseline characterization. Left: `hmmer` is dominated by conflict misses; `libquantum` is insensitive to associativity. Right: the stream buffer eliminates 97.7% of `libquantum` misses at $D = 1$ and then saturates; `hmmer` and `dealII` see negligible benefit.



(a) Total L2 requests vs. depth ($N = 1$).

(b) Effective L1D misses vs. stream count ($D = 1$).

Figure 2: Phase 1 bandwidth and multi-stream results. L2 traffic grows at $(D + 1) \times$ baseline for `hmmer` — nearly 9× at $D = 8$ — while effective misses barely move. Adding streams helps `libquantum` only marginally (already saturated at $N = 1$).

Three findings from Phase 1 motivate the Phase 2 design:

1. **Sequential workloads saturate at $D = 1$.** Because the simulator has no latency model, depth provides no additional benefit to hit rate beyond the first prefetch. In real hardware, depth would be needed to keep D lines simultaneously in flight to hide L2 round-trip latency.
2. **Irregular workloads suffer from aggressive prefetching.** `hmmer`'s L2 traffic grows 8.8× at $D = 8$ with no corresponding miss reduction.
3. **The functional simulator cannot distinguish latency effects.** To make speedup a meaningful metric, a latency model is required.

3 Phase 2: Adaptive Stream Buffer

3.1 Design Overview

Phase 2 re-implements the stream buffer following Palacharla and Kessler, with three additions over Phase 1: a two-level cache hierarchy with explicit hit costs, an adaptive prefetch-depth policy driven by a learned stream-length histogram, and a prefetch buffer that tracks in-flight lines and models ready vs. late hits.

Three policies are evaluated side-by-side against a no-prefetch baseline: **Off** (no prefetching), **NextLine** (always fetch the next sequential line on every L2 miss), and **Adaptive** (histogram-driven depth selection).

3.2 Cache Hierarchy and Latency Model

`SetAssocCache` implements set-associative LRU at configurable (size, associativity) for both L1D (4 KB, 1-way) and L2 (1 MB, 1-way), matching the Phase 1 `config-base`. The latency model assigns:

Event	Modeled cost (reference units)
L1D hit	1
L2 hit (L1D miss)	10
Memory access (L2 miss)	80
Prefetch latency	8

The stream buffer observes *L2 misses* (not L1D misses), matching the paper's placement. A prefetched line is inserted into both L1D and L2 on arrival; a subsequent demand hit to that line counts as an L1D hit at cost 1 rather than a memory access at cost 80.

3.3 Stream Tracking

Up to `stream_slots` active streams are tracked simultaneously. Each `StreamSlot` records: the most recently observed line, the inferred direction (+1, −1, or unknown), the current stream length, a `remaining_life` countdown, and the logical timestamp of the last update. On every L2-miss read, the tracker searches for a matching slot (adjacent line, consistent direction). If found, the slot is extended; if not, a new slot is allocated or the oldest is evicted.

3.4 Histogram-Driven Adaptive Depth

The adaptive policy maintains a cumulative histogram `lht_curr_[k]` counting how often observed streams have reached length k. Every `epoch_reads` L2 misses the epoch rolls over, replacing the current histogram with the next one. To choose the prefetch depth for a stream of current length ℓ, the policy walks the histogram forward:

```
depth = 0
probe = current_length
while probe < max_stream_length:
    if lht_curr_[probe] < 2 * lht_curr_[probe + 1]:
        depth++; probe++
    else:
        break
```

This strategy keeps prefetching as long as it is more likely than not that the stream continues at least one more line. This naturally produces shallow prefetching for `hmmer` (short or non-existent streams) and deep prefetching for `libquantum` (long sequential streams).

4 Experimental Methodology

Phase 1. 102 Pin configurations sweep L1D associativity (1–8-way), stream buffer depth (0–8), and stream count (1–16) across three benchmarks, capped at 5 billion retired instructions each. Metrics are effective L1D misses and total L2 requests.

Phase 2. 72 configurations sweep: 3 policies × 3 benchmarks × 2 stream slots $\{4, 8\}$ × 2 max prefetch depths $\{2, 8\}$ × 2 max stream lengths $\{8, 32\}$, capped at 5 billion instructions. Metrics are modeled-cycle speedup vs. the no-prefetch baseline (based on the accumulated costs from the latency model), prefetch accuracy (useful prefetches / issued prefetches), and prefetch coverage (read misses covered by useful prefetches / total read misses).

5 Results and Analysis

5.1 Speedup

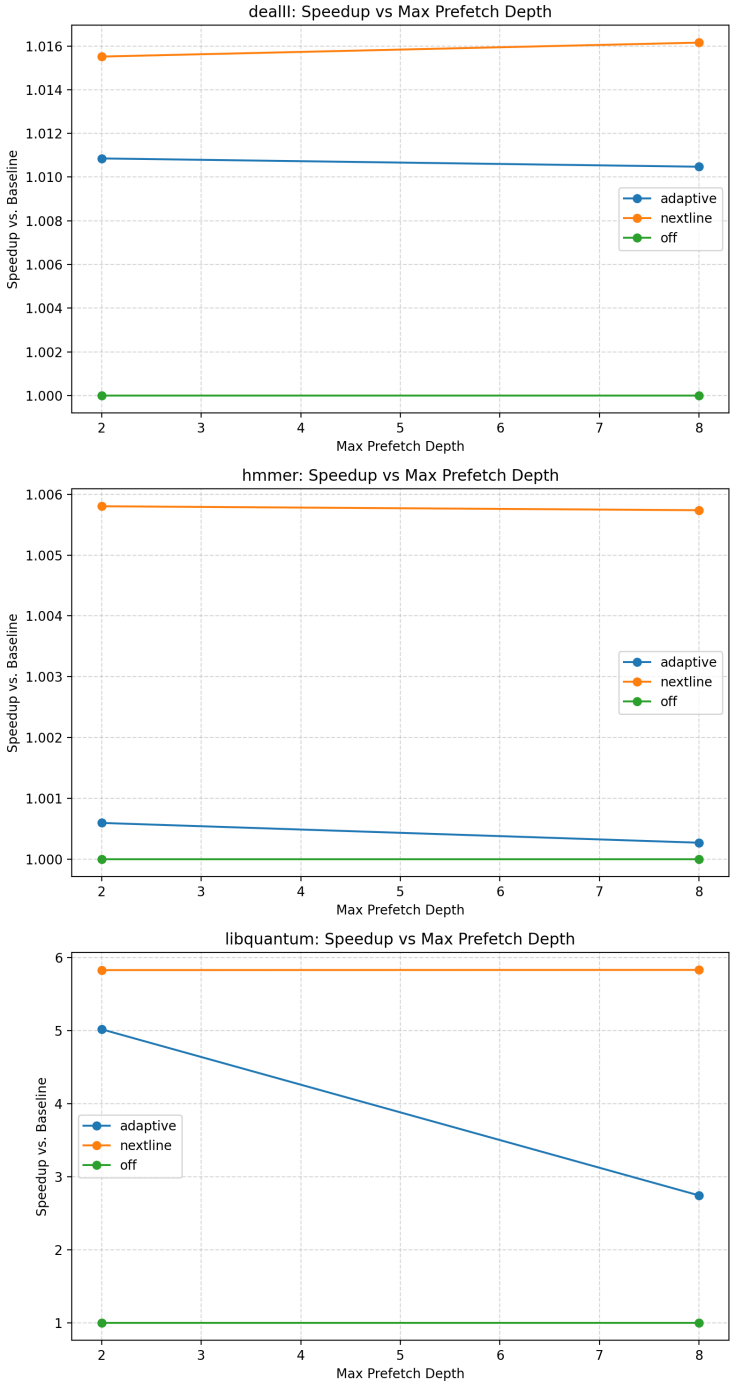


Figure 3: Speedup vs. baseline (no prefetch) for all three policies across benchmarks, as a function of max prefetch depth. For example, a depth of 8 means at most 8 lines are prefetched ahead of the current demand address at any moment.

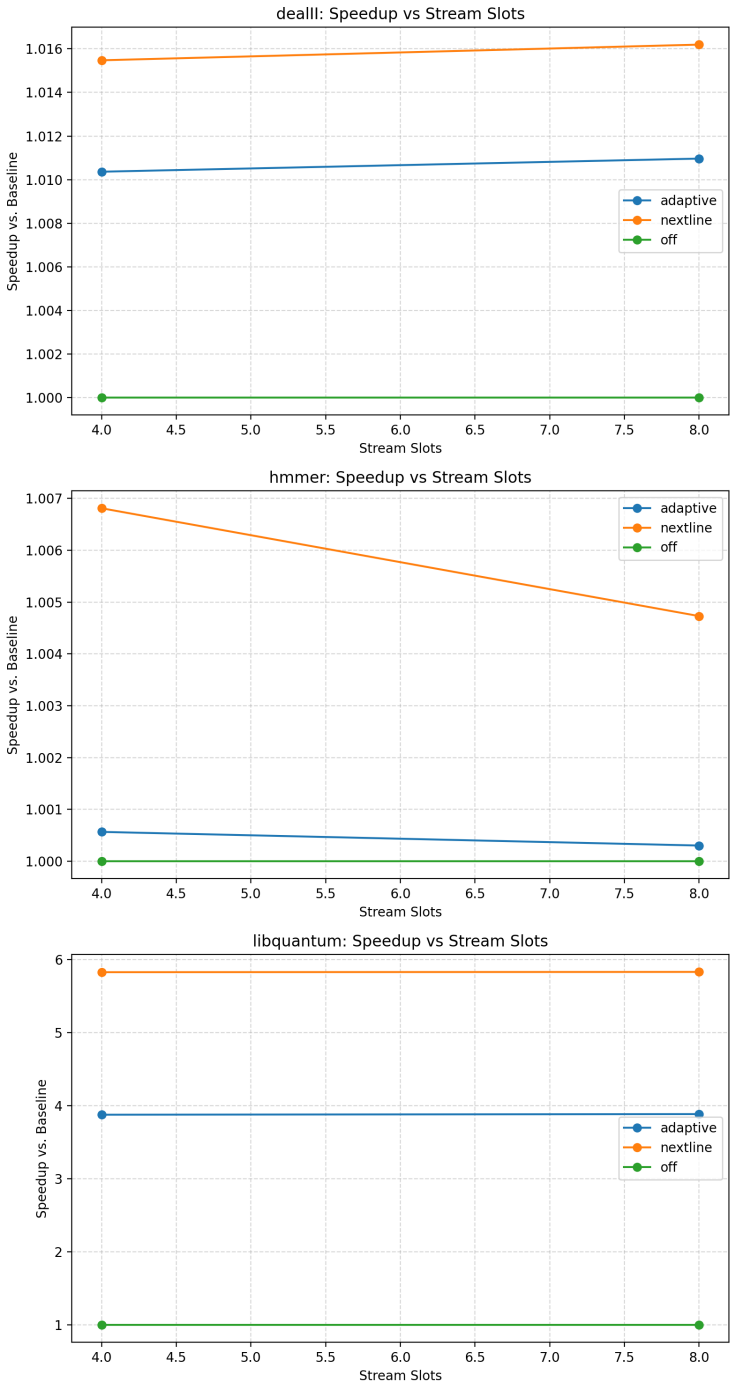


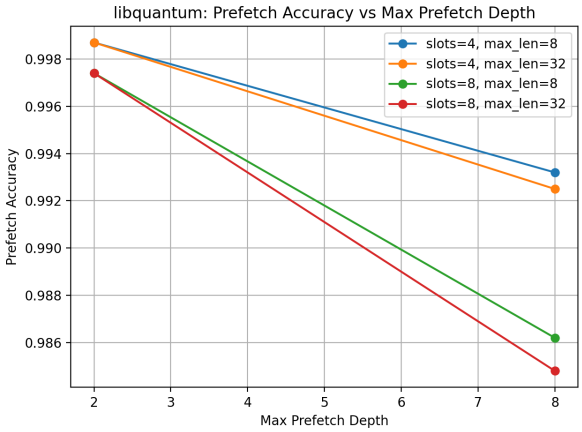
Figure 4: Speedup vs. number of stream slots.

dealII. NextLine achieves 1.016–1.018×; Adaptive 1.010–1.011×. The gap is small but consistent. The finite-element workload has enough sequential structure that both policies find useful prefetches, but the irregular sparse-matrix component dilutes the benefit.

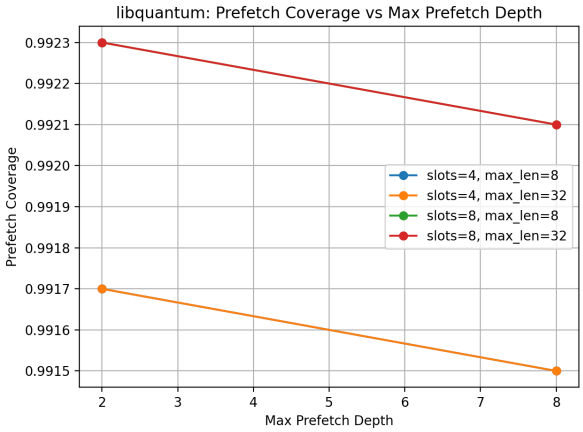
hmmer. Both NextLine and Adaptive provide negligible speedup ($\leq 1.009\times$). This aligns with Phase 1: the access pattern is fundamentally irregular, so any sequential prefetch is rarely useful.

libquantum. NextLine achieves a consistent **5.83**× speedup regardless of depth or slot count — confirming Phase 1's finding that a single look-ahead line is sufficient to hide latency for perfectly sequential access. Adaptive reaches **5.02**× at depth 2 (86% of next-line), then drops to 2.74× at depth 8. The degradation occurs because at high depth the histogram has not yet committed to "long stream" during warmup epochs, so the prefetcher oscillates between conservative and aggressive depths, accumulating late prefetch penalties. At depth 2 the learning cost is negligible and the policy stabilises quickly.

5.2 Accuracy and Coverage

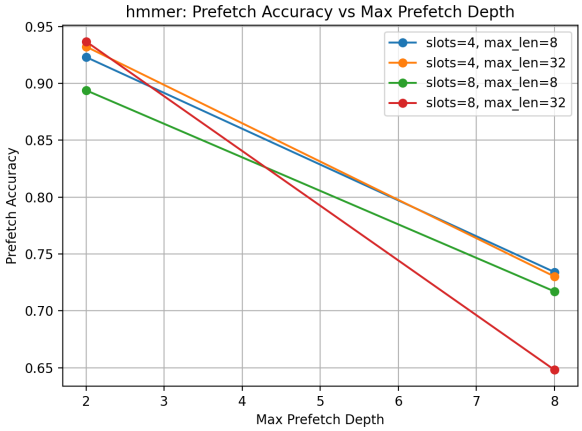


(a) `libquantum` accuracy.

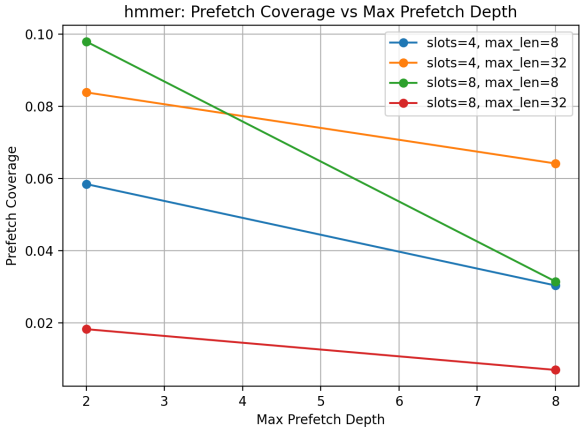


(b) `libquantum` coverage.

Figure 5: Prefetch accuracy and coverage vs. max prefetch depth for `libquantum`.

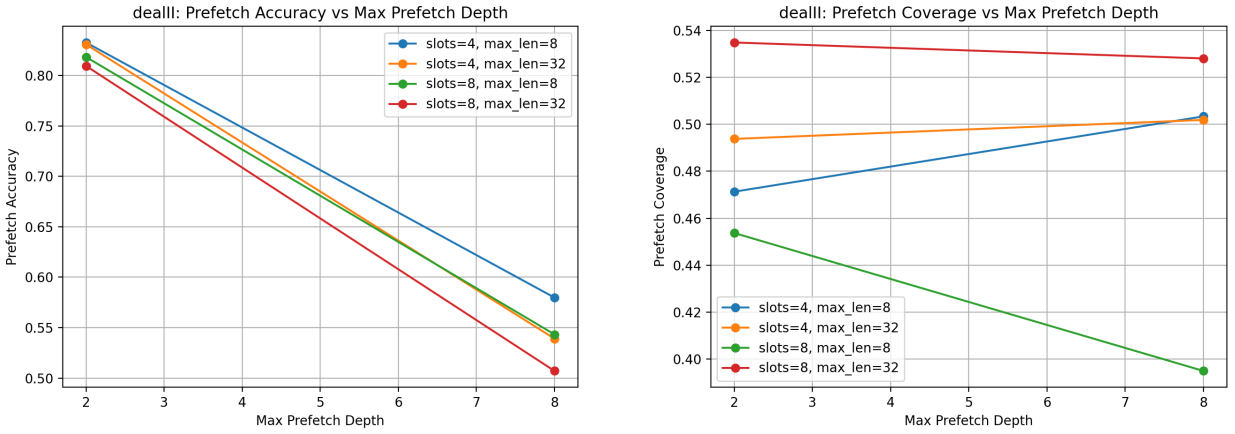


(a) `hmmer` accuracy.



(b) `hmmer` coverage.

Figure 6: Prefetch accuracy and coverage vs. max prefetch depth for `hmmer`.



(a) `dealII` accuracy.

(b) `dealII` coverage.

Figure 7: Prefetch accuracy and coverage vs. max prefetch depth for `dealII`.

		Speedup		Accuracy (%)		Coverage (%)	
Benchmark	Policy	d=2	d=8	d=2	d=8	d=2	d=8
	Off	1.00	1.00	—	—	—	—
`libquantum`	NextLine	5.83	5.83	99.4	99.4	99.2	99.2
	Adaptive	5.02	**2.74**	99.8	98.9	99.2	99.2
	Off	1.000	1.000	—	—	—	—
`hmmer`	NextLine	1.006	1.006	93.2	93.6	**78.2**	**76.8**
	Adaptive	1.001	1.000	92.2	**70.7**	**6.5**	**3.3**
	Off	1.000	1.000	—	—	—	—
`dealII`	NextLine	1.016	1.016	78.0	78.7	62.3	63.4
	Adaptive	1.011	1.011	**82.3**	**54.2**	48.8	48.2

Table 1: Mean speedup, prefetch accuracy, and prefetch coverage averaged over stream slot count and max stream length, at max prefetch depth d=2 and d=8. Off policy issues no prefetches so accuracy and coverage are undefined (—). **Bold** values indicate substantial divergence from the other policies or across depths.

libquantum. As shown in Table 1, all three policies achieve near-identical accuracy and coverage on `libquantum`. Adaptive reaches 99.8% accuracy and 99.2% coverage at d=2, matching NextLine's 99.4% and 99.2%. At d=8 adaptive accuracy dips slightly to 98.9% while coverage remains 99.2% — the small accuracy loss occurs because at higher depth the policy occasionally prefetches beyond where the stream ends before the slot expires. The configuration parameters (slot count, max stream length) have no measurable effect on either metric for this uniformly sequential workload.

hmmer. The most striking divergence in the table is `hmmer` coverage: NextLine achieves 78.2% at d=2 and 76.8% at d=8, while Adaptive achieves only 6.5% and 3.3% respectively. Both policies produce roughly the same negligible speedup (∼1.006× vs. ∼1.001×), but NextLine spends roughly 12× more memory bandwidth to reach the same outcome. Adaptive accuracy also degrades with depth — from 92.2% at d=2 to 70.7% at d=8 — because at higher depth caps the policy is occasionally induced to issue prefetches before the histogram has enough evidence that the stream

continues, resulting in more wasted fetches. The low coverage is the intended self-throttling behavior: the histogram has learned that `hmmer` streams are short, so it suppresses most prefetch candidates.

dealII. Table 1 shows a clear accuracy reversal across depths for `dealII`. At d=2, Adaptive accuracy is 82.3% versus NextLine's 78.0% — the histogram prevents prefetching in the irregular sparse-matrix regions where NextLine blindly issues fetches that go unused. At d=8 this advantage inverts sharply: Adaptive drops to 54.2% while NextLine holds steady at 78.7%. With a higher depth cap the policy is sometimes induced to prefetch aggressively before sufficient histogram evidence has accumulated, producing a large number of unused fetches. Coverage follows the opposite pattern: NextLine covers 62.3% of misses at d=2 versus Adaptive's 48.8%, reflecting that Adaptive is more selective about when it prefetches.

5.3 Hardware Cost vs. Performance

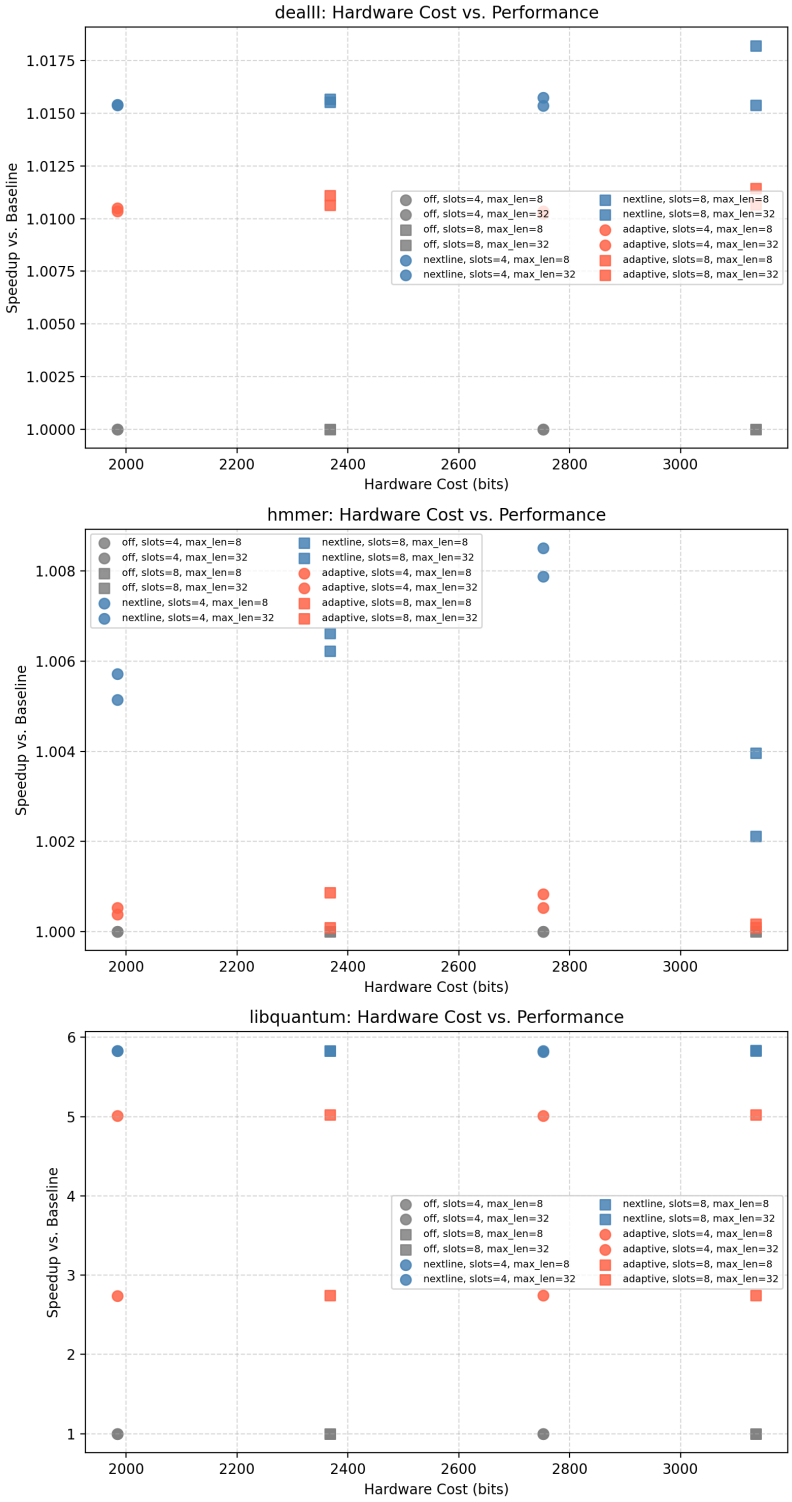


Figure 8: Hardware cost (bits) vs. speedup for all three policies. Cost accounts for stream slot storage, histogram tables, and prefetch buffer entries.

Hardware cost is estimated in bits as a function of stream slot count, max stream length (histogram table width), and max prefetch depth (prefetch buffer capacity). Across the configurations swept, cost ranges from ∼1984 to ∼3136 bits (∼250–400 bytes).

6 Discussion

The core trade-off. The adaptive policy sacrifices some peak speedup on perfectly sequential workloads in exchange for two benefits: (1) substantially higher prefetch accuracy on mixed workloads (`dealII` 83% vs. 78%), and (2) near-zero wasted bandwidth on irregular workloads (`hmmer` coverage 5–10% vs. 66–91% for next-line). In a system where L2 and memory bandwidth are shared resources, the bandwidth savings on irregular workloads can improve throughput for co-running threads even when the prefetcher provides no direct per-thread benefit.

Depth sensitivity. Phase 1 showed that in a functional simulator, depth only affects L2 traffic not hit rate. Phase 2 reveals the full picture with a latency model: at `max_prefetch_depth` = 2, adaptive achieves 5.02× on `libquantum`; at depth 8 it drops to 2.74×. This non-monotonic relationship arises because deeper depth enables the policy to be more aggressive once the histogram has learned the stream length distribution, but during the learning warmup phase deeper depth creates more late-prefetch penalties. The next-line policy, which does not consult a histogram, is immune to this warmup effect and maintains 5.83× across all depths.

7 Conclusions

This project demonstrates the complete design trajectory from Jouppi's original stream buffer to Palacharla and Kessler's adaptive extension. The principal conclusions are:

1. A fixed-depth FIFO stream buffer provides near-optimal miss reduction for purely sequential workloads at the minimum depth of one line. Deeper buffers improve latency hiding in real hardware but show no benefit in a functional simulator.
2. The same fixed-depth design causes catastrophic bandwidth waste on irregular workloads, generating up to 8.8× excess L2 traffic with negligible miss reduction. This motivates a throttling mechanism.
3. The histogram-driven adaptive policy self-throttles on irregular workloads (coverage 5–10% on `hmmer`) while preserving 86% of next-line's speedup on sequential workloads at `max_prefetch_depth` = 2.
4. Accuracy on mixed workloads (`dealII` 83% vs. 78%) is higher for adaptive than for next-line, confirming that selective prefetching outperforms always-on prefetching when the access pattern is heterogeneous.
5. Hardware cost (250–400 bytes) is modest and performance is not resource-limited: adding more bits does not bridge the remaining gap between adaptive and next-line on `libquantum`.

References

[1] Norman P. Jouppi. Improving direct-mapped cache performance by the addition of a small fully-associative cache and prefetch buffers. In *ISCA*, 1990.

[2] Subbarao Palacharla and Richard E. Kessler. Evaluating stream buffers as a secondary cache replacement. In *ISCA*, 1994.